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Systematic Review on Prediction and Monitoring of Climate Change

Anil Kumar | Umang Singh Chauhan | Mumukshu Pratap Singh | Pradeep Suman | Arvind Kumar

Environmental Science · International Journal of Recent Research and… · 10 December 2023

Climate change poses a severe and very bad threat to human society and global ecosystems, necessitating careful planning, monitoring, and forecasting. This paper **reviews spread** a range of monitoring… Expand

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A Statistical Investigation of how Slope affects a Wildfire’s Rate of Spread

Owen Goetz | Environmental Science · 2021

Modelling **wildfire spread** from data collected on various slopes and across di↵erent fuel types is an important way to understand how di↵erent parameters influence the speed at which these fires… Expand

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Using Drone Swarm to Stop Wildfire: A Predict-then-optimize Approach

Shijie Pan | Aoran Cheng | +4 authors | Zuo-Jun Max Shen

Environmental Science, Computer Science · arXiv.org · 25 November 2024

TLDR This work proposes a mixed-integer programming (MIP) model coupled with dynamic programming (DP) to enable efficient drone swarm task planning and uses chance-constrained robust optimization (CCRO) to ensure robust firefighting performances under varying situations. Expand

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Using cellular automata and multi-criteria evaluation to simulate the wildfire expansion in Prince George, British Co-lumbia

Shujian Jin | Qijian He | Tsz Ki Venus Heung | Environmental Science · Theoretical and Natural Science · 28 May 2024

Wildfires pose a critical and ongoing challenge in British Columbia, Canada, threatening human life, property, and natural ecosystems. To understand and predict the behavior of such fires, our study… Expand

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Assessment of Wildfire Risk on Transmission Assets

Dange Huang | B. Bagen | Environmental Science, Engineering · IEEE International Conference on Probabilistic… · 24 June 2024

The escalating threat of **wildfires** has been inflicting growing harm on power systems in recent years. Transmission lines, especially those supported by wooden poles, are susceptible to destruction by… Expand

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Forecasting and Surveillance of COVID-19 Spread Using Google Trends: Literature Review

Tobias Saegner | D. Austys | Medicine, Computer Science · International Journal of Environmental Research… · 29 September 2022

TLDR In most countries, the use of GT data showed to be beneficial for forecasting and surveillance of COVID-19 **spread**, and later, the search volumes reduced even though the

incidence peaked. Expand

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A review of the effects of wildfire smoke on the health and behavior of wildlife

O. V. Sanderfoot S. B. Bassing +4 authors B. Gardner Environmental Science · Environmental Research Letters · 19 October 2021

Climate change is intensifying global **wildfire** activity, and people and wildlife are increasingly exposed to hazardous air pollution during large-scale smoke events. Although **wildfire** smoke is… Expand

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Real-time fire segmentation via Active Contours for UAV integrated wildfire propagation prediction

F. Vivó M. Battipede P. Gili A. Yezzi E. Feron Eric N. Johnson

Environmental Science, Engineering · 8 January 2018

TLDR A robust fire segmentation algorithm for **wildfire** front tracking is presented, based on the solution of Partial Differential Equations (PDE) to model a time evolving curve and an efficient implementation of the Level Set method enables the algorithm to fulfil real time requirements. Expand

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Utilizing MODIS remote sensing and integrated data for forest fire spread modeling in the southwest region of Canada

H. Dastour Quazi K. Hassan Environmental Science · Environmental Research Communications · 31 January 2024

Accurate **prediction** of fire **spread** is considered crucial for facilitating effective fire management, enabling proactive planning, and efficient allocation of resources. This study places its focus on… Expand

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Combining MCTS and A3C for Prediction of Spatially Spreading Processes in Forest Wildfire Settings

Sriram Ganapathi Subramanian Mark Crowley Computer Science, Environmental Science · Canadian AI · 8 May 2018

TLDR A novel combination of Monte-Carlo Tree Search (MCTS) and A3C algorithms is introduced on an online simulator of a **wildfire**, on a pair of forest fires in Northern Alberta and on historical Saskatchewan fires previously compared by others to a physics-based simulator. Expand

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